Automation Script to Generate Zone/DoS traffic

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Tools Used	Python3 Hping3 Nmap
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- Purpose:
- Traffic Generation:
  - DOS-Protection:
  - Zone Protection:
- Configuration Template:

Purpose:

The intention of the tool is to enable quick simulations of respective Zone-Protection and DOS test cases during the below scenarios:

1. Customer Demos
2. SD-Security Zone/DoS Protection testing
3. Zone/DoS Protection from VSA Client
4. Quick Functional testing on new releases

The tool tries to cover all possible scenarios to demonstrate Zone-Protection & DOS related tests using Python. There will also be a Service Template which can be cloned/exported to any environments for re-usability.

Traffic Generation:

For Testing Zone Protection profile make sure the ingress interface belongs to a zone and respective Zone protection profile is mapped. And for DoS Protection profiles make sure the ingress and egress interfaces belongs to different zones or in case if both the interfaces belongs to same zone, make sure the interfaces belongs to different subnets.

Prerequisites:

Following tools needs to be installed on Ubuntu Desktop-18.04.

1. Python3
2. Hping3 – sudo apt-get install hping3
3. Nmap – sudo apt-get install nmap
4. VSA Client(as per use-case) – versa-sase-client-7.4.2-0-x86_64.deb

Ubuntu VSA Client download link: https://versanetworks.app.box.com/s/nvtbn8ty27q90clhxqhzb13kny7bg3t5/folder/147893195914

NOTE: Zone Protection does not work when attack traffic generated from VSA client device

Bug-id: http://bugzilla.versa-networks.com/show_bug.cgi?id=69499

- Traffic-generation script: **ZP_DoS_trafficgen.py**
- For generating traffic, copy the attached python file to Ubuntu-18.04 PC from where you'll generate traffic.
- Once copied, run the file from the Ubuntu Terminal using:

sudo python3 <file_name.py>

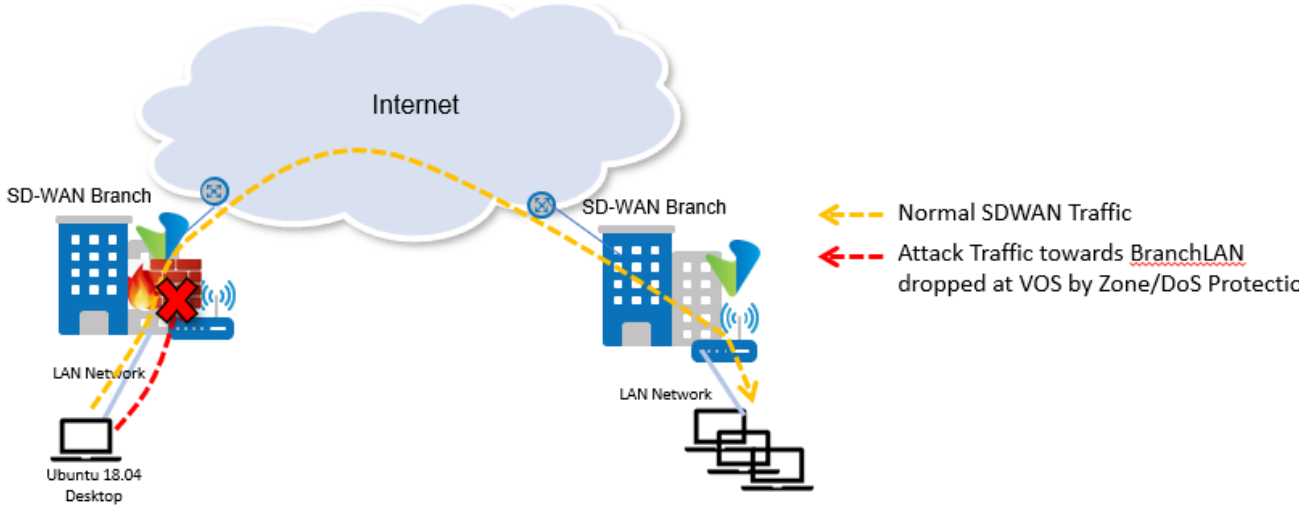
- This will prompt you to choose the traffic generation profile.

Example scenario used for this wiki section:

Here, for this script demonstration, we use an Ubuntu Desktop hosted in VOS Branch LAN network to simulate attack traffic generation. The VOS gateway is configured with Zone/DoS Protection profiles. Attacks here are simulated for SDWAN Overlay traffic, that is, anomalous packets are launched from Ubuntu machine (LAN) towards another SDWAN-Branch LAN network as shown in topology.

(Please refer the service template attached later in this section to view the Zone/DoS Protection rules)

The purpose is to use the script to easily generate any anomalous traffic from Ubuntu machine (LAN) to verify the Firewall rule actions configured in the VOS device.



DOS-Protection:

- Execute the python script & select **"1"** for DoSProtection traffic generation profile,

```
lab@lab:~/Downloads$ sudo python3 zp_dos-final.py

 Please choose the feature to be tested :

                  1.DoSProtection          2. ZoneProtection       3. END-SCRIPT
1
```

- Under DoS, we have the following attack profiles:

1. TCP Flood
2. UDP Flood
3. ICMP Flood
4. SCTP Flood
5. Other-IP Flood

```
 Please choose the traffic-gen profile :

                  1.TCP Flood      2.UDP Flood              3.ICMP Flood
                  4.SCTP Flood     5.Other-IP Flood         6.EXIT
```

- Select any of the flood profiles as required

For example:

- Selecting "1" will ask for, Target/Destination IP, TCP Destination Port and the duration of flood

```
Please choose the traffic-gen profile :

                1.TCP Flood      2.UDP Flood            3.ICMP Flood
                4.SCTP Flood     5.Other-IP Flood       6.EXIT
1

        Enter Target/Destination IP Address: 192.168.114.2

        Enter TCP Destination Port: 80

        Enter time duration (in sec) for which traffic is to be generated: 20
```

- Enter the required details & press ENTER to begin traffic generation for the specified IP/Port/Duration,

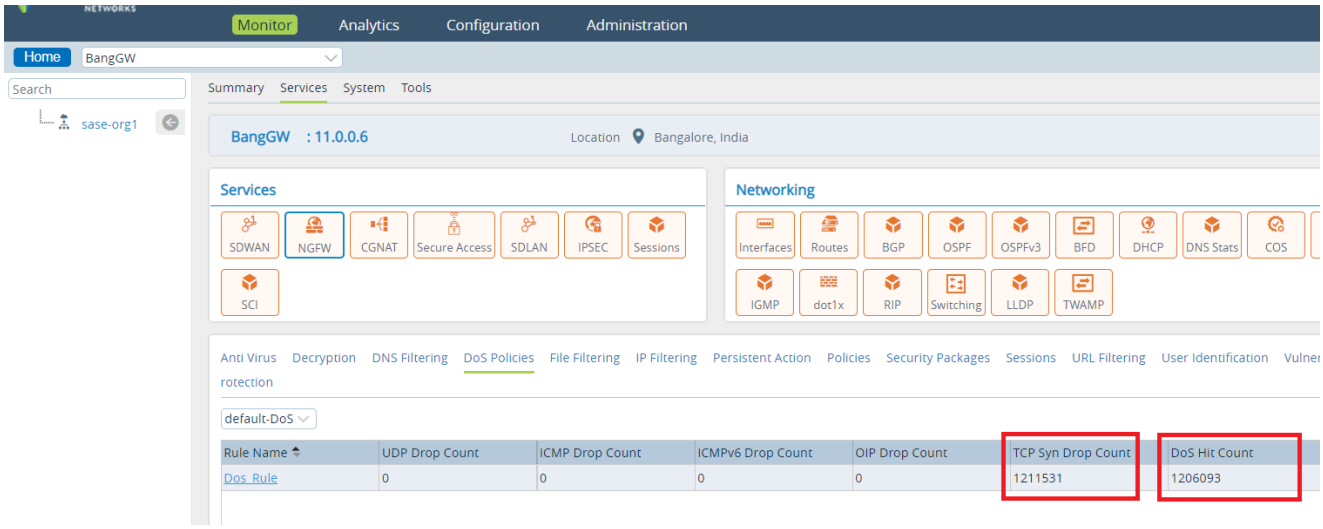
NOTE: Traffic will automatically terminate after the given duration

```
**********  Starting TCP SYN Flood  *********

Flood automatically terminates in 20 seconds

HPING 192.168.114.2 (eth1 192.168.114.2): S set, 40 headers + 0 data bytes

**********  Flooding Terminated  *********


--- 192.168.114.2 hping statistic ---
1121843 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

- **DoS-Policy verification:**

DoS rule hit-counts & the dropped TCP SYN counts on VOS device can be verified from Director GUI,

Go to the appliance & navigate to **Monitor -> Services -> SFW/NGFW -> DoS Policies** :



- The same can be viewed from VOS device cli using the command,

"show orgs org-services <Org-Name> security dos-policies rules dos-policy-stats"

```
admin@BangGW-cli> show orgs org-services sase-orgl security dos-policies rules
                        UDP    ICMP   ICMPV6  OIP    TCP SYN           SCTP   SESSION
              RULE      DROP   DROP   DROP    DROP   DROP     DOS HIT  DROP   DROP
NAME          NAME      COUNT  COUNT  COUNT   COUNT  COUNT    COUNT    COUNT  COUNT
-----------------------------------------------------------------------------------------
default-DoS   Dos_Rule  0      0      0       0      1325570  1315201  0      0
```

- Similarly any of the other DoS traffic-profiles can be used,

```
lab@lab:~/Downloads$ sudo python3 zp_dos-final.py

 Please choose the feature to be tested :

                 1.DoSProtection         2. ZoneProtection       3. END-SCRIPT
1

 Please choose the traffic-gen profile :

                 1.TCP Flood     2.UDP Flood             3.ICMP Flood
                 4.SCTP Flood    5.Other-IP Flood        6.EXIT
1
```

- Once done, we can either choose to EXIT from the script (or) continue with the traffic-gen

Zone Protection:

Under ZoneProtection, we have the below categorized traffic-gen profiles for the different types of attack /anomalous traffic that will be detected by VOS Zone Protection module

Scan	Flood	Packet-based Attack (TCP/UDP/IP/ICMP)
TCP Scan	TCP SYN Flood	ICMP Fragmentation
UDP Scan	UDP Flood	ICMP Ping Zero ID
Host-Sweep	ICMP Flood	Non-SYN TCP
	SCTP Flood	IP Spoofing
	Other-IP Flood	IP Fragmentation
		Record-Route
		Strict-source Routing
		Loose-source Routing
		Timestamp

Below are few examples to demonstrate how to use the python script on a LAN device for traffic generation

- ICMP Ping Zero ID
- Host-Sweep Scan
- Strict-source Routing

ICMP Ping Zero ID

- Run the script using python3 & enter "**2**" to view Zone Protection traffic profiles

```
swetha@ubuntu18:~/test-scripts$ sudo python3 zp_dos-final.py

 Please choose the feature to be tested :

                1.DoSProtection         2. ZoneProtection       3. END-SCRIPT
2

 Please choose the traffic-gen profile :

Scan:           1.TCP Scan              2.UDP Scan              3.HostSweep
Flood:          4.TCP Flood             5.UDP Flood             6.ICMP Flood            7.SCTP Flood            8.Other-IP Flo
Proto options:  9.ICMP Fragmention      10.ICMP Ping Zero ID    11.Non-SYN TCP
IP Options:     12.IP Spoofing          13.IP Fragmention       14.Record-Route
                15.Strict-src-Routing   16.Loose-src-Routing    17.Timestamp
                20.EXIT
```

- Input the value "**10**" for Ping Zero ID ICMP traffic & type the Target IP address & time duration (in seconds) in the prompt, press ENTER to initiate traffic-gen

```
                20.EXIT
10

        Enter Target/Destination IP Address: 192.168.51.2

        Enter time duration (in sec) for which traffic is to be generated: 10

 **********  Sending Ping Zero ID Packets *********

 Traffic automatically terminates in 10 seconds


 **********  Traffic Terminated  *********
```

NOTE: Traffic is automatically terminated after the specified duration & takes us back to the list of ZP traffic profiles.

Once done, we can either choose to EXIT from the script as shown below (or) continue with the traffic-gen

```
 **********  Traffic Terminated  *********


 Please choose the traffic-gen profile :

Scan:           1.TCP Scan              2.UDP Scan              3.HostSweep
Flood:          4.TCP Flood             5.UDP Flood             6.ICMP Flood            7.SCTP
Proto options:  9.ICMP Fragmention      10.ICMP Ping Zero ID    11.Non-SYN TCP
IP Options:     12.IP Spoofing          13.IP Fragmention       14.Record-Route
                15.Strict-src-Routing   16.Loose-src-Routing    17.Timestamp
                20.EXIT
20


 =======  Exiting Zone-Protection Traffic Profile  ==========


 Please choose the feature to be tested :

                1.DoSProtection         2. ZoneProtection       3. END-SCRIPT
3


 =======  EXITING TRAFFIC GENERATION  ==========

swetha@ubuntu18:~/test-scripts$
```

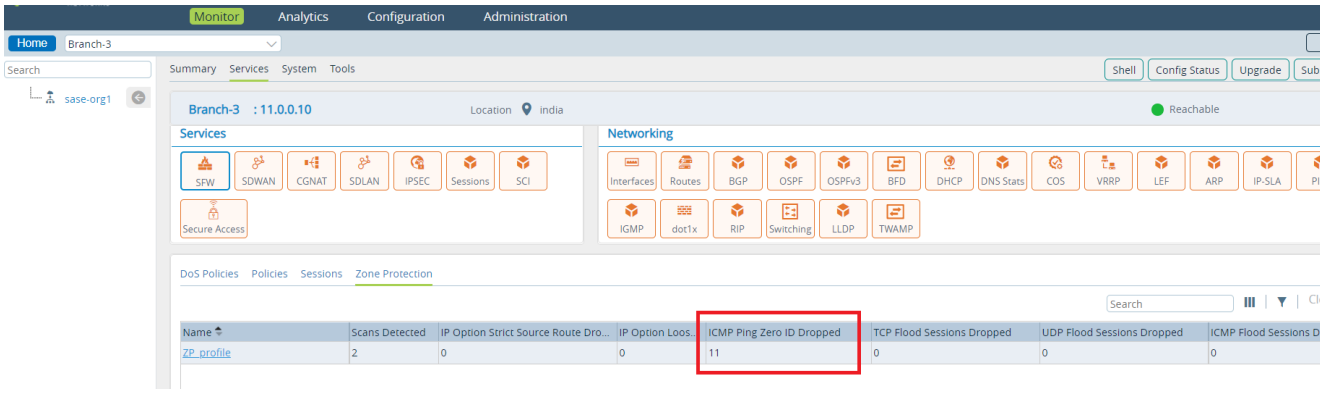
- **Verification:**

Use the below command on VOS device to check the number of anomalous data packets detected and dropped by the configured Zone Protection profile

"show orgs org-services <org-name> security profiles zone-protection zone-protection-statistics"

```
security profiles zone-protection zone-protection-statistics test
 scans-detected                          0
 ip-spoof-dropped                        0
 ip-fragment-dropped                     0
 ip-option-strict-source-route-dropped   0
 ip-option-loose-source-route-dropped    0
 ip-option-timestamp-dropped             0
 ip-option-record-route-dropped          0
 ip-option-security-id-dropped           0
 ip-option-stream-id-dropped             0
 ip-option-unknown-class-dropped         0
 ip-malform-dropped                      0
 tcp-non-syn-dropped                     0
 icmp-error-dropped                      0
 icmp-fragment-dropped                   0
 icmp-large-packet-dropped               0
 icmp-ping-zero-id-dropped               11
 log-alerts-sent                         0
 icmp-malform-dropped                    0
```

The same can be verified from Director GUI, go to the appliance & navigate to **Monitor -> Services -> SFW /NGFW -> Zone Protection** as shown:



Host-Sweep Scan

- As above, run the script using python3, enter "**2**" to get into Zone Protection traffic profiles
- Type "**3**" & press ENTER for Host-Sweep Scan. In the prompt, input the Target Subnet **x.x.x.x/x** to be scanned, the scan duration (in sec) & press ENTER to begin traffic-gen

```
 Please choose the traffic-gen profile :

Scan:            1.TCP Scan              2.UDP Scan              3.HostSweep
Flood:           4.TCP Flood             5.UDP Flood             6.ICMP Flood            7.SCTP Floo
Proto options:   9.ICMP Fragmention      10.ICMP Ping Zero ID    11.Non-SYN TCP
IP Options:      12.IP Spoofing          13.IP Fragmention       14.Record-Route
                 15.Strict-src-Routing   16.Loose-src-Routing    17.Timestamp
                 20.EXIT
3

         Enter Target/Destination Subnet x.x.x.x/x : 192.168.51.0/24

         Enter time duration (in sec) for which traffic is to be generated: 4

 **********  Starting HostSweep  *********

 Scan automatically terminates in 4 seconds


Starting Nmap 7.60 ( https://nmap.org ) at 2021-10-14 18:27 IST

 **********  HostSweep Terminated  *********
```

NOTE: Scan is automatically terminated by the script after the specified time

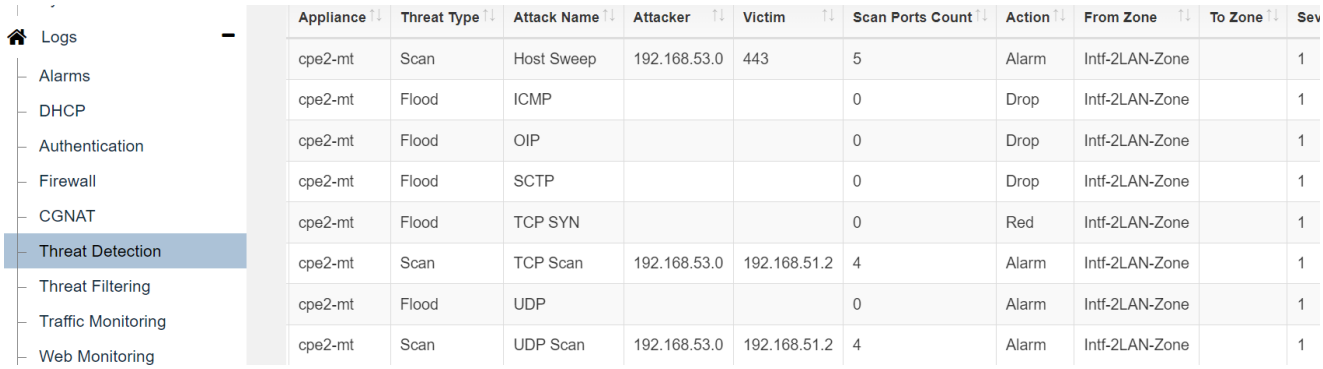
- **Verification:**

Use the below command on VOS device to check the number of anomalous data packets detected and dropped by the configured Zone Protection profile

"show orgs org-services <org-name> security profiles zone-protection zone-protection-statistics"

```
security profiles zone-protection zone-protection-statistics test
 scans-detected                        1
 ip-spoof-dropped                      0
 ip-fragment-dropped                   0
 ip-option-strict-source-route-dropped 0
 ip-option-loose-source-route-dropped  0
 ip-option-timestamp-dropped           0
 ip-option-record-route-dropped        0
 ip-option-security-id-dropped         0
 ip-option-stream-id-dropped           0
 ip-option-unknown-class-dropped       0
 ip-malform-dropped                    0
 tcp-non-syn-dropped                   0
 icmp-error-dropped                    0
 icmp-fragment-dropped                 0
 icmp-large-packet-dropped             0
 icmp-ping-zero-id-dropped             0
 log-alerts-sent                       2
 icmp-malform-dropped                  0
```

Additionally, the Scan & Flood attack traffic detected by VOS can also be verified from the Versa Analytics Dashboard under 'Threat Detection' as shown below (provided LEF is configured)



Appliance	Threat Type	Attack Name	Attacker	Victim	Scan Ports Count	Action	From Zone	To Zone	Sev
cpe2-mt	Scan	Host Sweep	192.168.53.0	443	5	Alarm	Intf-2LAN-Zone		1
cpe2-mt	Flood	ICMP			0	Drop	Intf-2LAN-Zone		1
cpe2-mt	Flood	OIP			0	Drop	Intf-2LAN-Zone		1
cpe2-mt	Flood	SCTP			0	Drop	Intf-2LAN-Zone		1
cpe2-mt	Flood	TCP SYN			0	Red	Intf-2LAN-Zone		1
cpe2-mt	Scan	TCP Scan	192.168.53.0	192.168.51.2	4	Alarm	Intf-2LAN-Zone		1
cpe2-mt	Flood	UDP			0	Alarm	Intf-2LAN-Zone		1
cpe2-mt	Scan	UDP Scan	192.168.53.0	192.168.51.2	4	Alarm	Intf-2LAN-Zone		1

Strict-source Routing

- Follow same procedure to enter into **ZoneProtection** section of the script & choose "**15**" from the list of Zone Protection traffic profiles as shown below
- Type the Destination IP address & time duration for which traffic is to be forwarded & press ENTER to begin traffic-gen

```
 Please choose the feature to be tested :

                 1.DoSProtection         2. ZoneProtection       3. END-SCRIPT
2

 Please choose the traffic-gen profile :

Scan:            1.TCP Scan              2.UDP Scan              3.HostSweep
Flood:           4.TCP Flood             5.UDP Flood             6.ICMP Flood            7.SCTP Flood            8.Other-IP Flo
Proto options:   9.ICMP Fragmention      10.ICMP Ping Zero ID    11.Non-SYN TCP
IP Options:      12.IP Spoofing          13.IP Fragmention       14.Record-Route
                 15.Strict-src-Routing   16.Loose-src-Routing    17.Timestamp
                 20.EXIT
15

         Enter Target/Destination IP Address: 192.168.51.2

         Enter time duration (in sec) for which traffic is to be generated: 3

 **********  Sending IP Packets (Strict-source Routing) *********

 Traffic automatically terminates in 3 seconds


 **********  Traffic Terminated  *********
```

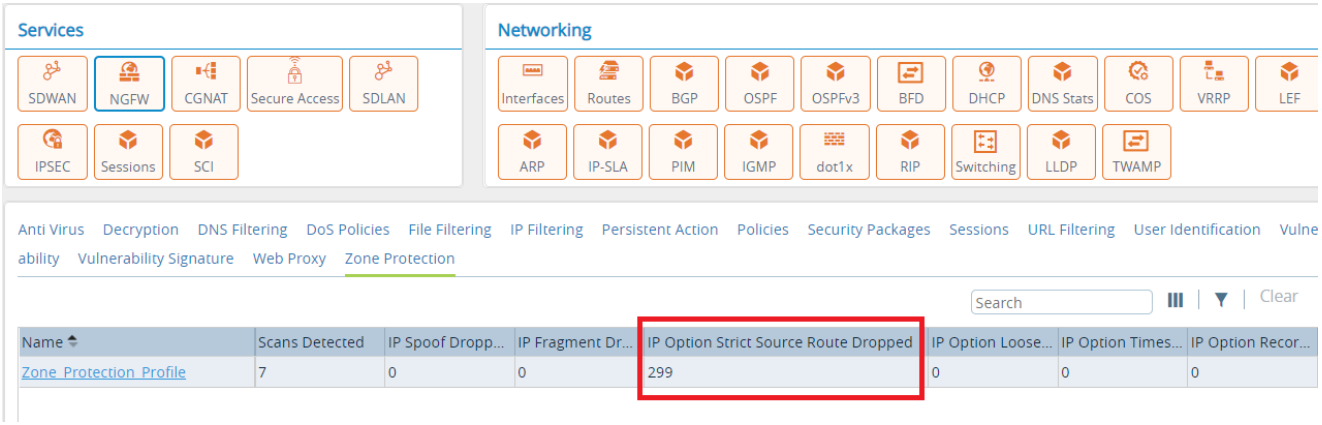
Verification:

Use the below command on VOS device to check the number of anomalous data packets detected and dropped by the configured Zone Protection profile

"show orgs org-services <org-name> security profiles zone-protection zone-protection-statistics"

```
security profiles zone-protection zone-protection-statistics test
 scans-detected                          0
 ip-spoof-dropped                        0
 ip-fragment-dropped                     0
 ip-option-strict-source-route-dropped 293
 ip-option-loose-source-route-dropped    0
 ip-option-timestamp-dropped             0
 ip-option-record-route-dropped          0
 ip-option-security-id-dropped           0
 ip-option-stream-id-dropped             0
 ip-option-unknown-class-dropped         0
 ip-malform-dropped                      0
 tcp-non-syn-dropped                     0
 icmp-error-dropped                      0
 icmp-fragment-dropped                   0
 icmp-large-packet-dropped               0
```

(OR) From Director GUI, got to the appliance & navigate to **Monitor -> Services -> SFW/NGFW -> Zone Protection :**



NOTE: Please feel free to try other available options and share feedback for any improvements.

Configuration Template:

You can use this template as reference or import this predefined service template into your test environment & modify as necessary

Service template (Type - Stateful) with Zone Protection & DOS Protection policies - SFW_Zone_and_DoS.cfg

NOTE:

- Zone Protection: Map the predefined profile to your respective zones as necessary
- DoS Protection: Map either the predefined 'Aggregate' or 'Classified' DoS profiles to the DoS policy rule based on use-case (by default, classified profile is configured)
- Map the Profiles matching your environment like mapping Zone Protection profiles to a desired Zone of your choice as per use-case. Similarly when using DOS profile, make sure to configure DoS rule match conditions as per your use-case.